In [2]:
```
import pandas as pd
import numpy as np

df = pd.DataFrame({
    "Mission_ID": [
        "M001", "M002", "M003", "M001", "M005",
        "M006", "M007", "M008", "M009", "M010",
        "M011", "M012", "M013", "M014", "M015",
        "M016", "M017", "M018", "M019", "M020"
    ],

    "Mission_Name": [
        "Mars Observer", "lunar explorer", "Voyager Probe",
        "Mars Observer", "Europa Clipper", "Chandrayaan-3",
        "SATURN EXPLORER", "Jupiter Orbiter", "Lunar Explorer",
        "Mars Rover", "Neptune Mission", "Solar Probe",
        "Comet Tracker", "Asteroid Survey", "Mars Rover",
        "Deep Space Probe", "Venus Orbiter", "Mercury Explorer",
        "ISS Research", "Pluto Explorer"
    ],

    "Observation_Date": [
        "2026-01-05", "2026-01-08", "2026-01-12",
        "2026-01-05", "2026-01-15", "2026-01-18",
        "2026-01-20", "2026-01-22", "2026-01-25",
        "2026-01-28", "2026-02-01", "2026-02-04",
        "2026-02-07", "2026-02-10", "2026-02-13",
        "2026-02-16", "2026-02-19", "2026-02-22",
        "2026-02-25", "invalid-date"
    ],

    "Region": [
        "Mars", "Moon", "Outer Solar System", "Mars",
        "Jupiter", "Moon", "Saturn", "Jupiter", "Moon",
        "Mars", "Outer Solar System", "Sun", "Asteroid Belt",
        "Asteroid Belt", "Mars", "Outer Solar System",
        "Venus", "Mercury", "Earth Orbit", "Outer Solar System"
    ],

    "Distance_Mkm": [
        225.4, 0.384, 24000, 225.4, 628,
        0.384, 1427, 778.5, 0.384, 225.4,
        4500, 0.7, 35.6, 12.8, 225.4,
        999999, 41.4, 77.3, 0.0004, "unknown"
    ],

    "Signal_Strength": [
        85, 92, 45, 85, 76, 95, 68, np.nan, 91, 82,
        39, 105, 72, 67, 88, 25, -10, 81, 89, 55
    ],

    "Data_Collected_GB": [
        12.5, 8.2, 2.8, 12.5, 15.4,
        10.1, 7.6, 18.3, np.nan, 11.7,
```

```
        3.2, 22.5, 6.4, 5.9, 1500,
        1.5, 9.8, 0, 14.2, 4.7
    ],

    "Status": [
        "Active", "Active", "Active", "Active", "In Transit",
        "Completed", "Active", "Active", "Active", "active",
        "In Transit", "Active", "Completed", "Unknown", "Active",
        "In Transit", "Completed", "Active", " Active ", "Completed"
    ]
})

df
```

Out[2]:

| | Mission_ID | Mission_Name | Observation_Date | Region | Distance_Mkm | Signal_Strength |
|---|---|---|---|---|---|---|
| **0** | M001 | Mars Observer | 2026-01-05 | Mars | 225.4 | 85.0 |
| **1** | M002 | lunar explorer | 2026-01-08 | Moon | 0.384 | 92.0 |
| **2** | M003 | Voyager Probe | 2026-01-12 | Outer Solar System | 24000 | 45.0 |
| **3** | M001 | Mars Observer | 2026-01-05 | Mars | 225.4 | 85.0 |
| **4** | M005 | Europa Clipper | 2026-01-15 | Jupiter | 628 | 76.0 |
| **5** | M006 | Chandrayaan-3 | 2026-01-18 | Moon | 0.384 | 95.0 |
| **6** | M007 | SATURN EXPLORER | 2026-01-20 | Saturn | 1427 | 68.0 |
| **7** | M008 | Jupiter Orbiter | 2026-01-22 | Jupiter | 778.5 | NaN |
| **8** | M009 | Lunar Explorer | 2026-01-25 | Moon | 0.384 | 91.0 |
| **9** | M010 | Mars Rover | 2026-01-28 | Mars | 225.4 | 82.0 |
| **10** | M011 | Neptune Mission | 2026-02-01 | Outer Solar System | 4500 | 39.0 |
| **11** | M012 | Solar Probe | 2026-02-04 | Sun | 0.7 | 105.0 |
| **12** | M013 | Comet Tracker | 2026-02-07 | Asteroid Belt | 35.6 | 72.0 |
| **13** | M014 | Asteroid Survey | 2026-02-10 | Asteroid Belt | 12.8 | 67.0 |
| **14** | M015 | Mars Rover | 2026-02-13 | Mars | 225.4 | 88.0 |
| **15** | M016 | Deep Space Probe | 2026-02-16 | Outer Solar System | 999999 | 25.0 |
| **16** | M017 | Venus Orbiter | 2026-02-19 | Venus | 41.4 | -10.0 |
| **17** | M018 | Mercury Explorer | 2026-02-22 | Mercury | 77.3 | 81.0 |
| **18** | M019 | ISS Research | 2026-02-25 | Earth Orbit | 0.0004 | 89.0 |
| **19** | M020 | Pluto Explorer | invalid-date | Outer Solar System | unknown | 55.0 |

In [4]:
```
df.head()
```

Out[4]:

| | Mission_ID | Mission_Name | Observation_Date | Region | Distance_Mkm | Signal_Strength | I |
|---|---|---|---|---|---|---|---|
| **0** | M001 | Mars Observer | 2026-01-05 | Mars | 225.4 | 85.0 | |
| **1** | M002 | lunar explorer | 2026-01-08 | Moon | 0.384 | 92.0 | |
| **2** | M003 | Voyager Probe | 2026-01-12 | Outer Solar System | 24000 | 45.0 | |
| **3** | M001 | Mars Observer | 2026-01-05 | Mars | 225.4 | 85.0 | |
| **4** | M005 | Europa Clipper | 2026-01-15 | Jupiter | 628 | 76.0 | |

In [5]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Mission_ID         20 non-null     str
 1   Mission_Name       20 non-null     str
 2   Observation_Date   20 non-null     str
 3   Region             20 non-null     str
 4   Distance_Mkm       20 non-null     object
 5   Signal_Strength    19 non-null     float64
 6   Data_Collected_GB  19 non-null     float64
 7   Status             20 non-null     str
dtypes: float64(2), object(1), str(5)
memory usage: 1.4+ KB
```

In [7]:
```
df.shape
```

Out[7]: (20, 8)

In [9]:
```
df.describe()
```

Out[9]:

| | Signal_Strength | Data_Collected_GB |
|---|---|---|
| **count** | 19.000000 | 19.000000 |
| **mean** | 70.000000 | 87.752632 |
| **std** | 28.347056 | 342.041307 |
| **min** | -10.000000 | 0.000000 |
| **25%** | 61.000000 | 5.300000 |
| **50%** | 81.000000 | 9.800000 |
| **75%** | 88.500000 | 13.350000 |
| **max** | 105.000000 | 1500.000000 |

In [11]:
```
df.isnull().sum()
```

Out[11]:
```
Mission_ID           0
Mission_Name         0
Observation_Date     0
Region               0
Distance_Mkm         0
Signal_Strength      1
Data_Collected_GB    1
Status               0
dtype: int64
```

In [13]:
```
df.isna().sum()
```

Out[13]:
```
Mission_ID           0
Mission_Name         0
Observation_Date     0
Region               0
Distance_Mkm         0
Signal_Strength      1
Data_Collected_GB    1
Status               0
dtype: int64
```

In [17]:
```
df["Signal_Strength"] = df["Signal_Strength"].fillna(0)
df["Signal_Strength"]
```

Out[17]:
```
0      85.0
1      92.0
2      45.0
3      85.0
4      76.0
5      95.0
6      68.0
7       0.0
8      91.0
9      82.0
10     39.0
11    105.0
12     72.0
13     67.0
14     88.0
15     25.0
16    -10.0
17     81.0
18     89.0
19     55.0
Name: Signal_Strength, dtype: float64
```

In [18]:
```
df["Data_Collected_GB"] = df["Data_Collected_GB"].fillna(
    df["Data_Collected_GB"].median()
)
```

In [21]:
```
df["Data_Collected_GB"]
```

Out[21]:
```
0       12.5
1        8.2
2        2.8
3       12.5
4       15.4
5       10.1
6        7.6
7       18.3
8        9.8
9       11.7
10       3.2
11      22.5
12       6.4
13       5.9
14    1500.0
15       1.5
16       9.8
17       0.0
18      14.2
19       4.7
Name: Data_Collected_GB, dtype: float64
```

In [24]:
```
df_without_missing = df.dropna()
print(df_without_missing.shape)
```

```
(20, 8)
```

In [26]:
```
df["Mission_Name"] = df["Mission_Name"].str.strip().str.title()
df["Status"] = df["Status"].str.strip().str.title()
```

In [28]:
```
df
```

Out[28]:

| | Mission_ID | Mission_Name | Observation_Date | Region | Distance_Mkm | Signal_Strength |
|---|---|---|---|---|---|---|
| **0** | M001 | Mars Observer | 2026-01-05 | Mars | 225.4 | 85.0 |
| **1** | M002 | Lunar Explorer | 2026-01-08 | Moon | 0.384 | 92.0 |
| **2** | M003 | Voyager Probe | 2026-01-12 | Outer Solar System | 24000 | 45.0 |
| **3** | M001 | Mars Observer | 2026-01-05 | Mars | 225.4 | 85.0 |
| **4** | M005 | Europa Clipper | 2026-01-15 | Jupiter | 628 | 76.0 |
| **5** | M006 | Chandrayaan-3 | 2026-01-18 | Moon | 0.384 | 95.0 |
| **6** | M007 | Saturn Explorer | 2026-01-20 | Saturn | 1427 | 68.0 |
| **7** | M008 | Jupiter Orbiter | 2026-01-22 | Jupiter | 778.5 | 0.0 |
| **8** | M009 | Lunar Explorer | 2026-01-25 | Moon | 0.384 | 91.0 |
| **9** | M010 | Mars Rover | 2026-01-28 | Mars | 225.4 | 82.0 |
| **10** | M011 | Neptune Mission | 2026-02-01 | Outer Solar System | 4500 | 39.0 |
| **11** | M012 | Solar Probe | 2026-02-04 | Sun | 0.7 | 105.0 |
| **12** | M013 | Comet Tracker | 2026-02-07 | Asteroid Belt | 35.6 | 72.0 |
| **13** | M014 | Asteroid Survey | 2026-02-10 | Asteroid Belt | 12.8 | 67.0 |
| **14** | M015 | Mars Rover | 2026-02-13 | Mars | 225.4 | 88.0 |
| **15** | M016 | Deep Space Probe | 2026-02-16 | Outer Solar System | 999999 | 25.0 |
| **16** | M017 | Venus Orbiter | 2026-02-19 | Venus | 41.4 | -10.0 |
| **17** | M018 | Mercury Explorer | 2026-02-22 | Mercury | 77.3 | 81.0 |
| **18** | M019 | Iss Research | 2026-02-25 | Earth Orbit | 0.0004 | 89.0 |
| **19** | M020 | Pluto Explorer | invalid-date | Outer Solar System | unknown | 55.0 |

In [30]:
```
df[["Mission_Name", "Region", "Status"]].head(10)
```

Out[30]:

| | Mission_Name | Region | Status |
|---|---|---|---|
| **0** | Mars Observer | Mars | Active |
| **1** | Lunar Explorer | Moon | Active |
| **2** | Voyager Probe | Outer Solar System | Active |
| **3** | Mars Observer | Mars | Active |
| **4** | Europa Clipper | Jupiter | In Transit |
| **5** | Chandrayaan-3 | Moon | Completed |
| **6** | Saturn Explorer | Saturn | Active |
| **7** | Jupiter Orbiter | Jupiter | Active |
| **8** | Lunar Explorer | Moon | Active |
| **9** | Mars Rover | Mars | Active |

In [32]:
```
df["Distance_Mkm"] = pd.to_numeric(
    df["Distance_Mkm"], errors="coerce"
)
```

In [34]:
```
df["Distance_Mkm"]
```

Out[34]:
```
0        225.4000
1          0.3840
2      24000.0000
3        225.4000
4        628.0000
5          0.3840
6       1427.0000
7        778.5000
8          0.3840
9        225.4000
10      4500.0000
11         0.7000
12        35.6000
13        12.8000
14       225.4000
15    999999.0000
16        41.4000
17        77.3000
18         0.0004
19            NaN
Name: Distance_Mkm, dtype: float64
```

In [36]:
```
df["Observation_Date"] = pd.to_datetime(
    df["Observation_Date"], errors="coerce"
)
```

In [38]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 20 entries, 0 to 19
Data columns (total 8 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Mission_ID         20 non-null     str
 1   Mission_Name       20 non-null     str
 2   Observation_Date   19 non-null     datetime64[us]
 3   Region             20 non-null     str
 4   Distance_Mkm       19 non-null     float64
 5   Signal_Strength    20 non-null     float64
 6   Data_Collected_GB  20 non-null     float64
 7   Status             20 non-null     str
dtypes: datetime64[us](1), float64(3), str(4)
memory usage: 1.4 KB
```

In [40]:
```
Q1 = df["Data_Collected_GB"].quantile(0.25)
Q3 = df["Data_Collected_GB"].quantile(0.75)

IQR = Q3 - Q1

lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

print("Q1:", Q1)
print("Q3:", Q3)
print("IQR:", IQR)
print("Lower bound:", lower_bound)
print("Upper bound:", upper_bound)
```

```
Q1: 5.6000000000000005
Q3: 12.925
IQR: 7.325
Lower bound: -5.3875
Upper bound: 23.9125
```

In [42]:
```
outliers = df[
    (df["Data_Collected_GB"] < lower_bound) |
    (df["Data_Collected_GB"] > upper_bound)
]

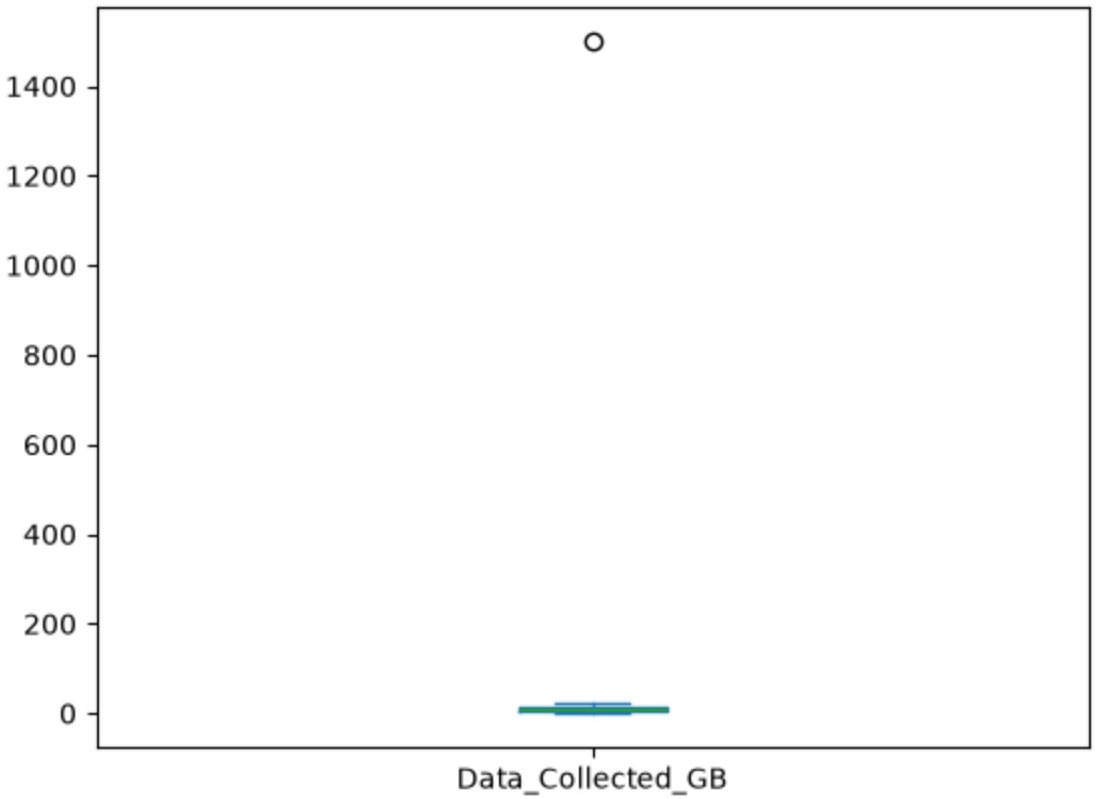
display(outliers)
```

| | Mission_ID | Mission_Name | Observation_Date | Region | Distance_Mkm | Signal_Strength | D |
|---|---|---|---|---|---|---|---|
| **14** | M015 | Mars Rover | 2026-02-13 | Mars | 225.4 | 88.0 | |

In [47]:
```
df["Data_Collected_GB"].plot(kind="box")
```

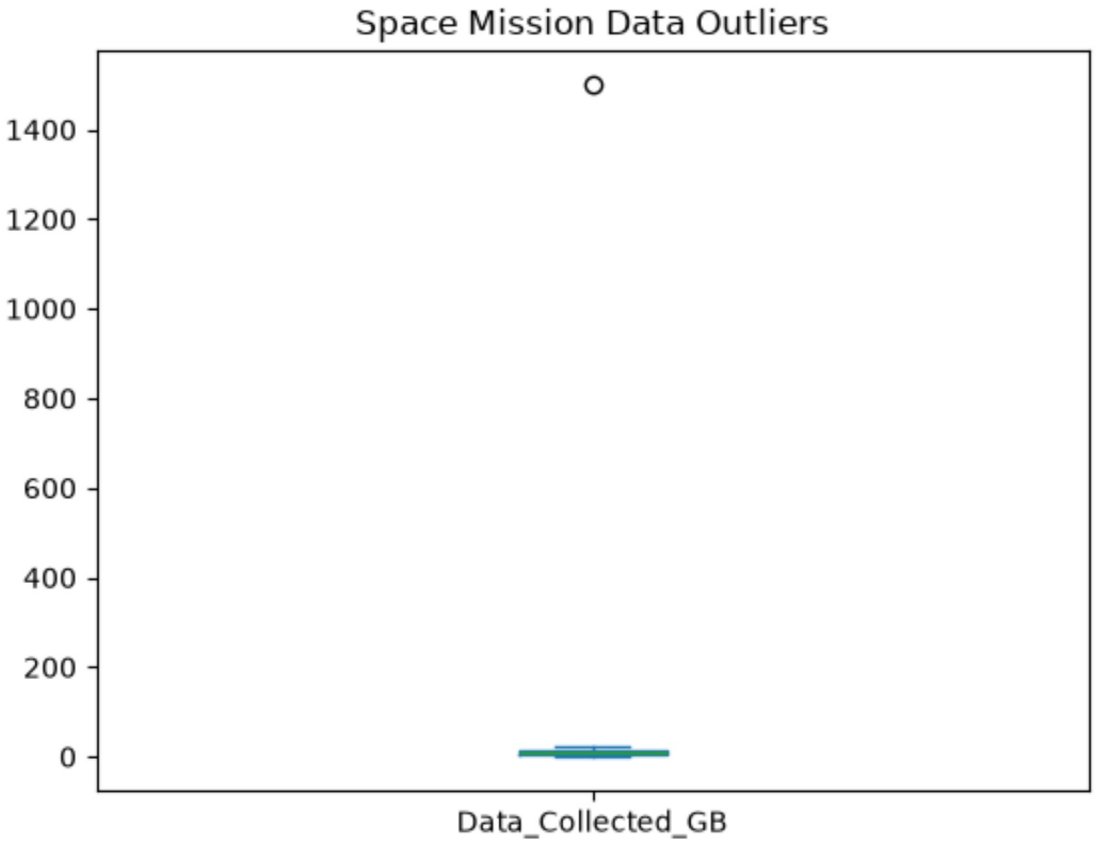
```
Matplotlib is building the font cache; this may take a moment.
```

Out[47]: <Axes: >




In [49]:
```
df["Data_Collected_GB"].plot(kind="box", title="Space Mission Data Outliers")
```

Out[49]: <Axes: title={'center': 'Space Mission Data Outliers'}>



In [52]:
```
df["Observation_Year"] = df["Observation_Date"].dt.year
df["Observation_Year"]
```

Out[52]:
```
0     2026.0
1     2026.0
2     2026.0
3     2026.0
4     2026.0
5     2026.0
6     2026.0
7     2026.0
8     2026.0
9     2026.0
10    2026.0
11    2026.0
12    2026.0
13    2026.0
14    2026.0
15    2026.0
16    2026.0
17    2026.0
18    2026.0
19       NaN
Name: Observation_Year, dtype: float64
```

In [55]:
```
df["Observation_Month"] = df["Observation_Date"].dt.month
df[["Observation_Year","Observation_Month"]]
```

Out[55]:

| | Observation_Year | Observation_Month |
|---|---|---|
| **0** | 2026.0 | 1.0 |
| **1** | 2026.0 | 1.0 |
| **2** | 2026.0 | 1.0 |
| **3** | 2026.0 | 1.0 |
| **4** | 2026.0 | 1.0 |
| **5** | 2026.0 | 1.0 |
| **6** | 2026.0 | 1.0 |
| **7** | 2026.0 | 1.0 |
| **8** | 2026.0 | 1.0 |
| **9** | 2026.0 | 1.0 |
| **10** | 2026.0 | 2.0 |
| **11** | 2026.0 | 2.0 |
| **12** | 2026.0 | 2.0 |
| **13** | 2026.0 | 2.0 |
| **14** | 2026.0 | 2.0 |
| **15** | 2026.0 | 2.0 |
| **16** | 2026.0 | 2.0 |
| **17** | 2026.0 | 2.0 |
| **18** | 2026.0 | 2.0 |
| **19** | NaN | NaN |

In [57]:
```
def classify_distance(distance):
    if pd.isna(distance):
        return "Unknown"
    elif distance < 1:
        return "Nearby"
    elif distance < 1000:
        return "Medium Range"
    else:
        return "Deep Space"

df["Distance_Category"] = df["Distance_Mkm"].apply(classify_distance)
```

In [59]:
```
df[["Mission_Name", "Distance_Mkm", "Distance_Category"]]
```

Out[59]:

| | Mission_Name | Distance_Mkm | Distance_Category |
|---|---|---|---|
| **0** | Mars Observer | 225.4000 | Medium Range |
| **1** | Lunar Explorer | 0.3840 | Nearby |
| **2** | Voyager Probe | 24000.0000 | Deep Space |
| **3** | Mars Observer | 225.4000 | Medium Range |
| **4** | Europa Clipper | 628.0000 | Medium Range |
| **5** | Chandrayaan-3 | 0.3840 | Nearby |
| **6** | Saturn Explorer | 1427.0000 | Deep Space |
| **7** | Jupiter Orbiter | 778.5000 | Medium Range |
| **8** | Lunar Explorer | 0.3840 | Nearby |
| **9** | Mars Rover | 225.4000 | Medium Range |
| **10** | Neptune Mission | 4500.0000 | Deep Space |
| **11** | Solar Probe | 0.7000 | Nearby |
| **12** | Comet Tracker | 35.6000 | Medium Range |
| **13** | Asteroid Survey | 12.8000 | Medium Range |
| **14** | Mars Rover | 225.4000 | Medium Range |
| **15** | Deep Space Probe | 999999.0000 | Deep Space |
| **16** | Venus Orbiter | 41.4000 | Medium Range |
| **17** | Mercury Explorer | 77.3000 | Medium Range |
| **18** | Iss Research | 0.0004 | Nearby |
| **19** | Pluto Explorer | NaN | Unknown |

In [62]:
```
df.loc[df["Distance_Mkm"] == "unknown", "Distance_Mkm"] = np.nan
```

In [64]:
```
df["Observation_Date"] = pd.to_datetime(
    df["Observation_Date"], errors="coerce"
)
```

In [66]:
```
df = df.dropna()
```

In [68]:
```
df.isnull().sum()
```

Out[68]:
```
Mission_ID           0
Mission_Name         0
Observation_Date     0
Region               0
Distance_Mkm         0
Signal_Strength      0
Data_Collected_GB    0
Status               0
Observation_Year     0
Observation_Month    0
Distance_Category    0
dtype: int64
```

In [70]:
```
df.shape
```

Out[70]:
```
(19, 11)
```

In [72]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 19 entries, 0 to 18
Data columns (total 11 columns):
 #   Column             Non-Null Count  Dtype
---  ------             --------------  -----
 0   Mission_ID         19 non-null     str
 1   Mission_Name       19 non-null     str
 2   Observation_Date   19 non-null     datetime64[us]
 3   Region             19 non-null     str
 4   Distance_Mkm       19 non-null     float64
 5   Signal_Strength    19 non-null     float64
 6   Data_Collected_GB  19 non-null     float64
 7   Status             19 non-null     str
 8   Observation_Year   19 non-null     float64
 9   Observation_Month  19 non-null     float64
 10  Distance_Category  19 non-null     str
dtypes: datetime64[us](1), float64(5), str(5)
memory usage: 1.8 KB
```

In [73]:
```
df.to_csv("space_missions_cleaned.csv", index=False)
```